THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sub-atomic particle discovered by J. J. Thomson was the: 1

(a) nucleus **(b)** neutron **(c)** proton **(d)** electron

2. The atomic number of an element is equal to the number of: 1

(a) protons **(b)** valence electrons only

(c) nucleons **(d)** neutrons

3. The valency of an element with 7 valence electrons is: 1

(a) 1 **(b)** 7 **(c)** 8 **(d)** 6

4. The charge on an electron is: 1

(a) variable **(b)** positive **(c)** zero **(d)** negative

5. The mass number of an atom equals the number of: 1

(a) neutrons plus electrons **(b)** protons only

(c) protons plus electrons **(d)** protons plus neutrons

6. An atom with a complete outermost shell is generally: 1

(a) chemically unreactive **(b)** radioactive

(c) very reactive **(d)** positive

7. The nucleus of an atom was discovered by: 1

(a) Thomson **(b)** Dalton **(c)** Bohr **(d)** Rutherford

8. An atom has 11 protons and 12 neutrons. Its mass number is: 1

(a) 34 **(b)** 12 **(c)** 23 **(d)** 11

9. Isotopes are atoms of the same element with the same: 1

(a) atomic number but different mass numbers **(b)** mass number but different atomic numbers

(c) number of nucleons **(d)** number of neutrons

10. The neutron was discovered by: 1

(a) Rutherford **(b)** Thomson **(c)** Bohr **(d)** James Chadwick

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State two limitations of Thomson's model of the atom. 2

12. Calculate the number of protons, neutrons and electrons in $^{23}_{11}\mathrm{Na}$. 2

13. State two uses of isotopes. 2

14. What did Rutherford conclude from the gold foil experiment? State two conclusions. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the gold foil experiment: set-up, observations and the conclusion. 3

16. Differentiate between isotopes and isobars (three points). 3

17. Explain the three isotopes of hydrogen with their composition. 3

OR

Explain how Chadwick's discovery of the neutron explained the atomic mass of atoms like helium. 3

18. State three drawbacks of Rutherford's model that were removed by Bohr's model. 3

OR

An ion has 10 electrons and 12 protons. Name the ion with its charge and electronic configuration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe Thomson's, Rutherford's and Bohr's models of the atom with one drawback of each of the first two. 5

OR

Explain how the atomic number and the mass number identify an atom. Complete the table for ${}^{39}_{19}\mathrm{K}$, ${}^{32}_{16}\mathrm{S}$ and ${}^{19}_{9}\mathrm{F}$. 5

20. Describe Rutherford's gold foil experiment with the observations and the inferences for each. 5

OR

Explain the formation of the ions Na^{+}, Mg^{2+}, Cl^{-} and O^{2-} in terms of electrons and noble gas configurations. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND